8.03SC Problem Set 5

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Problem 5.3

Consider a string of mass density ρ under tension T. A pulse is traveling on the string, causing a vertical displacement given by:

$$y(x,t) = y_0 e^{\left[-\frac{1}{2}\left(\frac{(x-vt)}{\sigma}\right)^2\right]}$$

Energy in the wave is a sum of kinetic energy of the moving string and the potential energy of the string deformation integrated over the whole length of the string.

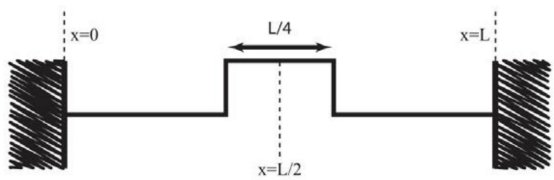


Figure 1: A string

a. For the displacement given above, find an expression for the kinetic energy density of the string at some time t and sketch the result as function of x.

b. Find an expression for the potential energy density of the string. Compare it with the kinetic energy density.

c. Find an expression for the total energy of the pulse in terms of T, y_0 and σ. Check the units. Recall that the energy should vary as a square of the amplitude. (Hint: You may need to use Gaussian integral $\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{1}{2a}\sqrt{\frac{\pi}{a}}$)

d. Find an expression for the power (Watts) flowing across the point $x = x_0$ as a function of time. Sketch the result.

a.

The kinetic energy density is

$$u_K = \frac{1}{2}\rho\left(\frac{\partial y}{\partial t}\right)^2$$

$$\frac{\partial y}{\partial t} = y_0 e^{-\frac{1}{2}\left(\frac{x-vt}{\sigma}\right)^2}\left(\frac{v(x-vt)}{\sigma^2}\right)$$

Hence,

$$\boxed{u_K(x,t) = \frac{1}{2}\rho y_0^2 \frac{v^2(x-vt)^2}{\sigma^4} \exp\left[-\left(\frac{x-vt}{\sigma}\right)^2\right]}$$

which is a Gaussian envelope multiplied by $(x-vt)^2$

b.

Potential energy density is

$$u_U = \frac{1}{2}T\left(\frac{\partial y}{\partial x}\right)^2$$

$$\frac{\partial y}{\partial x} = -y_0 e^{-\frac{1}{2}\left(\frac{x-vt}{\sigma}\right)^2}\left(\frac{x-vt}{\sigma^2}\right)$$

Therefore,

$$\boxed{u_U(x,t) = \frac{1}{2}T y_0^2 \frac{(x-vt)^2}{\sigma^4} \exp\left[-\left(\frac{x-vt}{\sigma}\right)^2\right]}$$

Using the wave relation

$$v^2 = \frac{T}{\rho}$$

we obtain

$$\boxed{u_U = u_K}$$

Thus kinetic and potential energy densities are equal.

c.

Total energy:

$$E = \int_{-\infty}^{\infty} (u_K + u_U)\, dx = 2\int_{-\infty}^{\infty} u_K\, dx$$

$$\implies E = \rho y_0^2 \frac{v^2}{\sigma^4}\int_{-\infty}^{\infty} (x-vt)^2 e^{-\left(\frac{x-vt}{\sigma}\right)^2} dx$$

$$\implies E = \rho y_0^2 \frac{v^2}{\sigma}\int_{-\infty}^{\infty} u^2 e^{-u^2}\, du$$

$$\implies E = \frac{\sqrt{\pi}}{2}\rho v^2 \frac{y_0^2}{\sigma}$$

Using $T = \rho v^2$:

$$\boxed{E = \frac{\sqrt{\pi}}{2}\frac{Ty_0^2}{\sigma}}$$

which correctly scales as y_0^2.

d.

Power transmitted along a string is

$$P(x,t) = -T\frac{\partial y}{\partial x}\frac{\partial y}{\partial t}$$

$$\implies P(x,t) = Ty_0^2\frac{v(x-vt)^2}{\sigma^4}\exp\left[-\left(\frac{x-vt}{\sigma}\right)^2\right]$$

Hence at $x = x_0$,

$$\boxed{P(x_0,t) = Ty_0^2\frac{v(x_0-vt)^2}{\sigma^4}\exp\left[-\left(\frac{x_0-vt}{\sigma}\right)^2\right]}$$

which represents a localized pulse of power moving with speed v.

Problem 5.4

Consider a string with linear density ρ_L split into two pieces. The two halves are attached to a massless ring which slides vertically without friction on a rod at $x = 0$ as shown in Figure 2. One of the two halves is stretched in the negative x direction with tension T_L, the other is stretched in the positive x direction with tension T_R. Note that the vertical rod is necessary to balance the horizontal forces on the massless string arising from the two string with different tensions.

Assume that there is a traveling wave of amplitude A coming from the negative x direction.

a. Assuming the incident wave has this functional form: $\psi(x,t) = \alpha\cos(kx - \omega t)$, write the general expression for the string displacement $\psi(x,t)$ on both sides of the ring.

b. Write the expressions for boundary conditions at $x = 0$. (Hint: since the ring is massless, the total force on the ring is 0)

c. Find the reflection (R) and transmission (T) coefficients for this system.

d. Show that the energy is conserved at the junction.

e. What are the reflection and transmission coefficients in the following three cases: $T_R = T_L$, $T_R \gg T_L$, and $T_R \ll T_L$? Does your result make sense?

a.

Incident wave:

$$\psi_i = A\cos(k_L x - \omega t)$$

The reflected and transmitted waves give

$$\boxed{\psi_L(x,t) = A\cos(k_L x - \omega t) + B\cos(k_L x + \omega t) \qquad x < 0}$$

$$\boxed{\psi_R(x,t) = C\cos(k_R x - \omega t) \qquad x > 0}$$

where

$$\boxed{k_L = \frac{\omega}{v_L} \quad k_R = \frac{\omega}{v_R} \quad v_{L,R} = \sqrt{\frac{T_{L,R}}{\rho_L}}}$$

b.

Continuity of displacement and Zero net force on massless ring needs to be satisfied

(i) Continuity of displacement

$$\psi_L(0,t) = \psi_R(0,t)$$

$$\implies \boxed{A + B = C}$$

(ii) Zero net force on massless ring

Vertical forces must cancel:

$$T_L \frac{\partial \psi_L}{\partial x} = T_R \frac{\partial \psi_R}{\partial x} \quad \text{at } x = 0$$

$$\implies \boxed{T_L k_L (A - B) = T_R k_R C}$$

c.

Solving

$$A + B = C, \qquad T_L k_L (A - B) = T_R k_R C$$

Using $k = \omega/v$ and $v = \sqrt{T/\rho_L}$ gives

$$Tk = \omega\sqrt{\rho_L T}$$

Define impedance

$$Z = \sqrt{\rho_L T}$$

Then

$$\frac{B}{A} = \frac{Z_L - Z_R}{Z_L + Z_R} \qquad \frac{C}{A} = \frac{2Z_L}{Z_L + Z_R}$$

Hence

$$\boxed{R = \left(\frac{Z_L - Z_R}{Z_L + Z_R}\right)^2}$$

$$\boxed{T = \frac{4Z_L Z_R}{(Z_L + Z_R)^2}}$$

d.

Power carried by a wave:

$$P = \frac{1}{2} Z\omega^2 A^2$$

Thus

$$P_i = P_r + P_t$$

becomes

$$Z_L A^2 = Z_L B^2 + Z_R C^2$$

which is satisfied by the above R and T, proving energy conservation.

e.

- $T_R = T_L \Rightarrow Z_R = Z_L$:

  $$\boxed{R = 0, \quad T = 1}$$

  (no reflection)

- $T_R \gg T_L \Rightarrow Z_R \gg Z_L$:

  $$\boxed{r \to -1}$$

  (reflection with sign inversion)

- $T_R \ll T_L \Rightarrow Z_R \ll Z_L$:

  $$\boxed{r \to 1}$$

  (reflection without inversion)

All limits agree with physical expectations.

Problem 5.5

A string of mass density ρ_L under tension T is attached by a massless ring to a wire covered with a viscous grease. The ring experiences a vertical drag force $F_y = -b\frac{\partial y}{\partial t}$ when the end of the string moves.

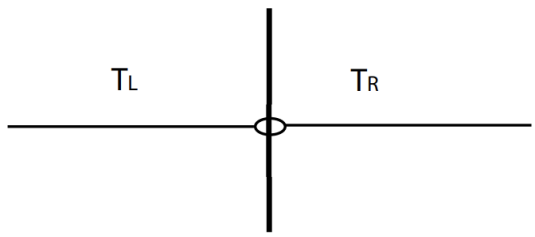
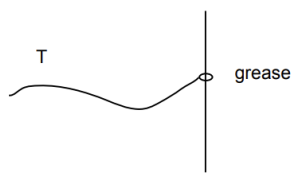


Figure 2: Two string halves with a ring in between



a. Use Newton's Law applied to the ring to find the boundary condition at the end of the string as a relation between the partial derivatives of $y(x,t)$ at that point. (Hint: there are three forces acting on the ring: string tension, normal force from the wire and the vertical drag force)

b. Show that the boundary condition is satisfied by an incident pulse $f(t-\frac{x}{v})$ and reflected pulse $g(t+\frac{x}{v})$. Find g in terms of f.

c. Show that your result has the proper behavior in the limits $b \to 0$ (free slip) and $b \to \infty$ (clamped string).

a.

Apply Newton's law to the massless ring. The vertical forces are:

- vertical component of string tension,
- normal force from the rod (horizontal only),
- viscous drag force $-b\,\partial_t y$.

Since the ring is massless, net vertical force is zero:

$$T\frac{\partial y}{\partial x}\Big|_{x=0} - b\,\frac{\partial y}{\partial t}\Big|_{x=0} = 0$$

Hence the boundary condition is

$$\boxed{T\,\partial_x y(0,t) = b\,\partial_t y(0,t)}$$

b.

Let the displacement be

$$y(x,t) = f\Big(t - \frac{x}{v}\Big) + g\Big(t + \frac{x}{v}\Big) \qquad v = \sqrt{\frac{T}{\rho_L}}$$

Derivatives:

$$\partial_x y = -\frac{1}{v} f' + \frac{1}{v} g' \qquad \partial_t y = f' + g'$$

Apply boundary condition at $x = 0$:

$$T\left(-\frac{1}{v} f' + \frac{1}{v} g'\right) = b(f' + g')$$

Rearranging,

$$(T - bv)g' = (T + bv)f'$$

Thus

$$\boxed{g'(t) = \frac{T + bv}{T - bv} f'(t)}$$

and integrating,

$$\boxed{g(t) = \frac{T + bv}{T - bv} f(t)}$$

The integration constant vanishes since localized pulses must disappear far from the interaction region.

c.

Free slip & Clamped string

(i) Free slip ($b \to 0$):

$$g = f$$

(no inversion; free end reflection).

(ii) Clamped string ($b \to \infty$):

$$g = -f$$

(reflection with inversion).

Both limits agree with physical expectations.